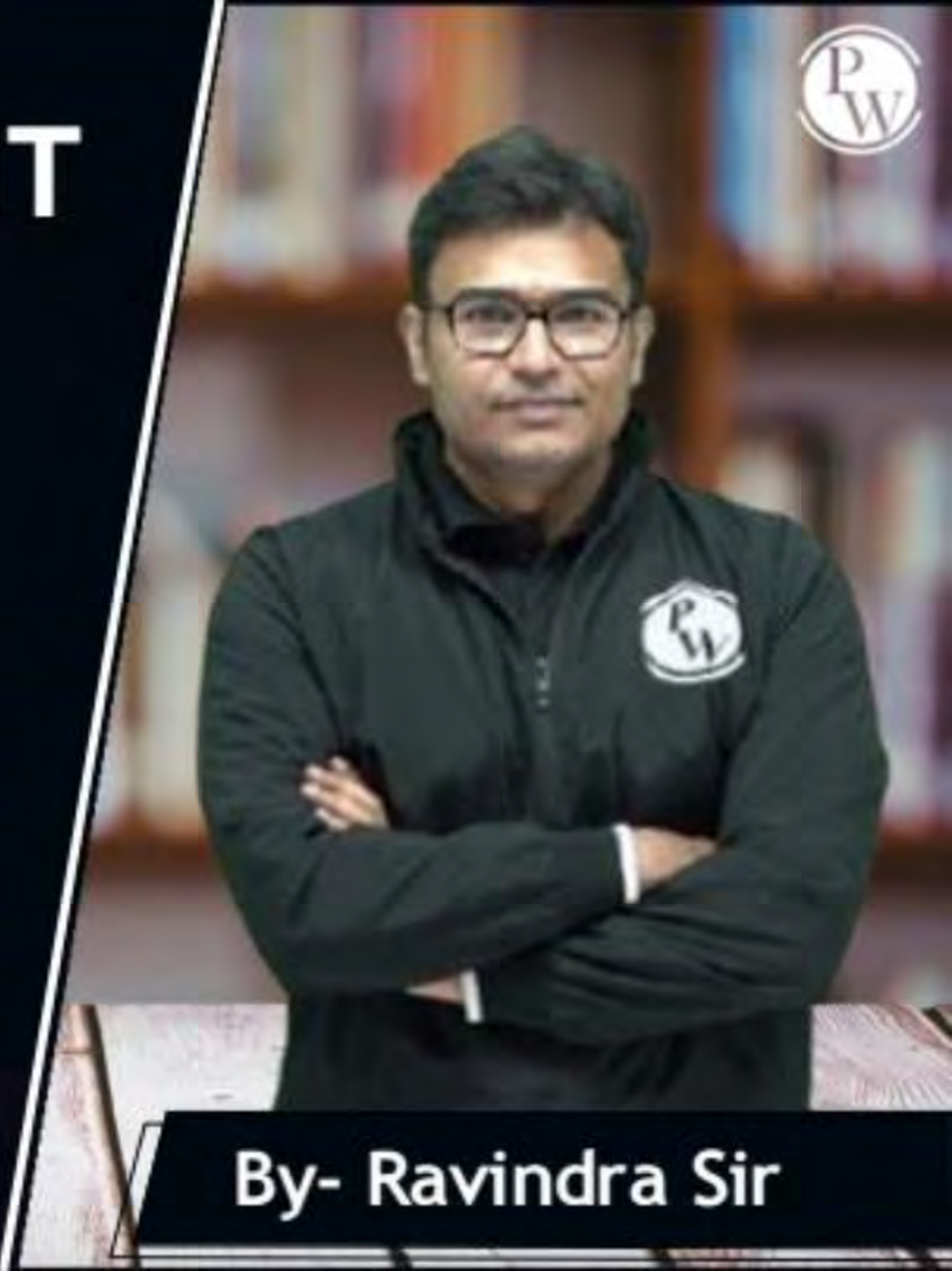
Computer Science & IT

COMPILER DESIGN
(CD)

Parsers

Lecture No. 14



By- Ravindra Sir

Recap of Previous Lecture



- Topic — LR(0), SLR(1)
- Topic
- Topic

Topics to be Covered



- Topic
- Topic
- Topic

CLR(1) LALR(1)

Join Telegram

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Inspiring Stories : Veeramangai Kuyili

Veeramangai Kuyili – The Woman Who Became a Human Torch

Background: Kuyili was a trusted soldier of queen Velu Nachiyar from Tamil Nadu in the 1700s.

Struggles: She worked without praise and lived in a time when women were rarely seen in battle.

Achievements: During one attack, she covered her body in ghee, set herself on fire, and jumped into the British armory. Her act became a key turning point in the war.

Impact: If she could light herself on fire to save her queen, you can light up that answer sheet with focus



Inspiring Stories : B. N. Jagatap

B. N. Jagatap

Background: Grew up in a modest family in rural Maharashtra with no strong science background at home.

Education: PhD in Nuclear Magnetic Resonance (NMR) from IISc Bangalore.

Career Achievements: Improved NMR methods to study molecules in great detail, helping drug design and materials research.

Impact: Made high-end science tools more effective for Indian research labs and universities.



Inspiring Stories : Mula Gabharu

Mula Gabharu – The Queen Who Raised Women Soldiers

Background: Mula Gabharu was the daughter of an Ahom king in Assam in the early 1500s.

Struggles: Her husband fell fighting an invading army. Her people were helpless and losing hope.

Achievements: She trained women warriors, led them into battle with courage, and confronted the enemy herself before being martyred.

Impact: If she could turn grief into action and lead women into battle, you can turn doubt into study and lead your rank up.



Inspiring Stories : Veeramangai Kuyili

K. J. Rao

Background: Born in a small town in Karnataka with very limited means. Had to work hard to afford studies.

Education: PhD in Solid State Chemistry from IISc Bangalore.

Career Achievements: Developed new ceramic materials that can store energy and work in extreme heat.

Impact: Helped India make better energy storage systems and heat-resistant tools for industry.

CLR(1)/LR(1) and LALR(1):-

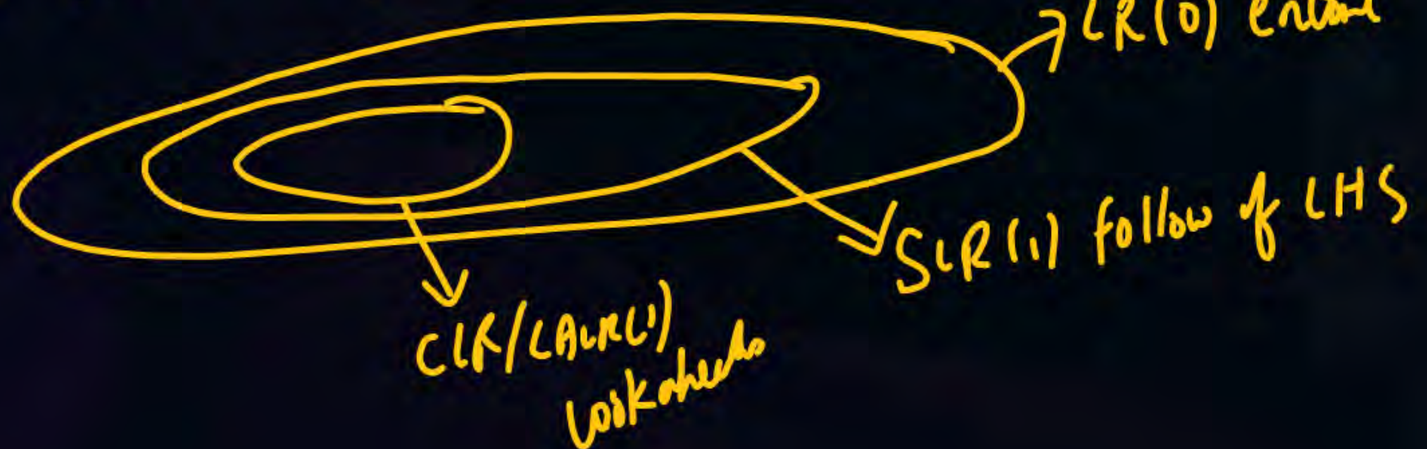
Among all the parsers CLR(1) is most powerful parser.

Parsing algorithm for all LR parser is same, only parsing table changes.

In LR(0), reduce moves are placed in the entire row.

In SLR(1), reduce moves are placed only in the follow(LHS)

In CLR(1) and LALR(1), reduce moves will be placed only under lookaheads



To Construct LALR(1) and CLR(1), we use LR(1) items

LR(1) item = LR(0) item + lookahead

Ex.- $\underbrace{S \to aA.}_{LR(0)}, \underbrace{a/b}_{\text{lookahead}}$ (we have to place reduce move under a, b)

Closure (I) where 'I' is a Set of LR(1) items :-

I is a set of LR(1) items like $[A \to \alpha . \beta B, \$], [C \to a., a]$

The function closure takes as i/p a set of LR(1) items and produces a set of LR(1) items as o/p. Following 2 rules are followed

1) Initially add every item from i/p to o/p
2) If $A \to \alpha . B\beta, (\$)$ is in I where B is a non terminal and $B \to \gamma$ is the rule for 'B', then add $B \to .\gamma$, $First(\beta\$)$ to closure (I). Repeat this for every newly added item.

Goto (I, x) :-

Given the set of all items of the form $[A \to \alpha . X\beta, a]$ in I

goto (I, x) = closure ($[A \to \alpha X . \beta, a]$)

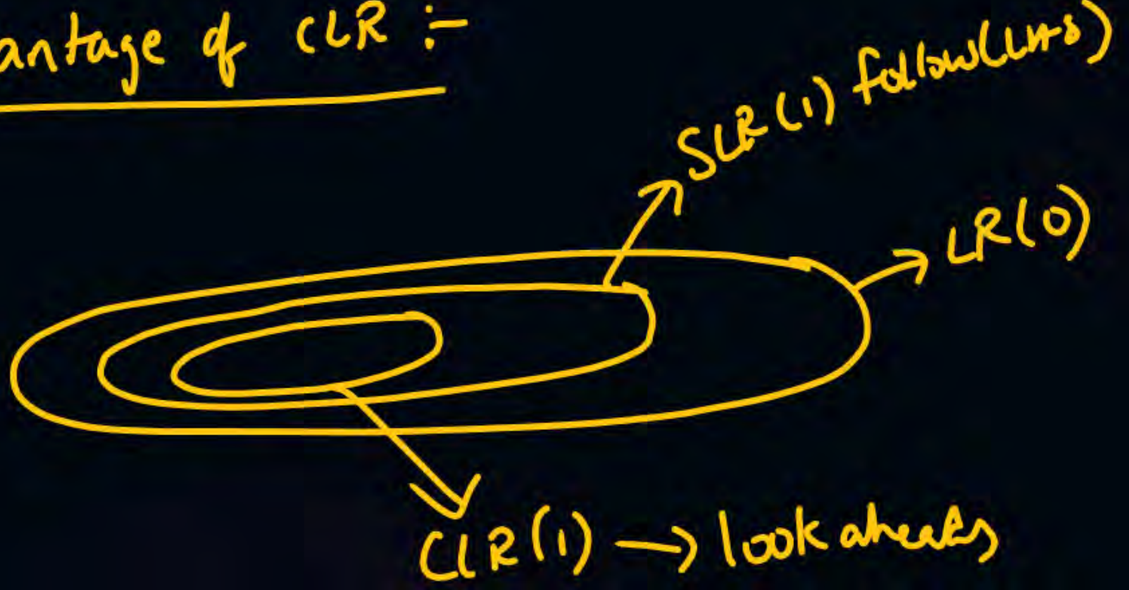
The goto function involves two steps

1) Find transition
2) Apply closure

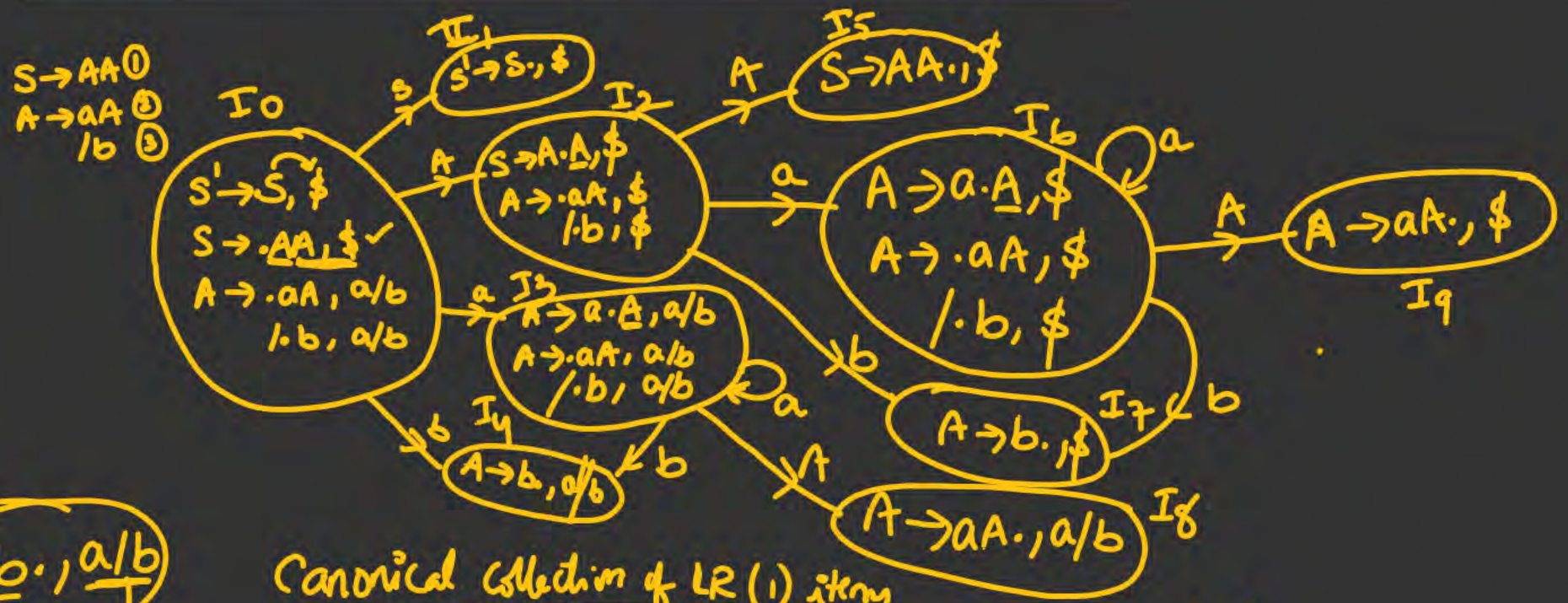
→ while finding transition, lookahead remains same

→ while finding closure, lookahead may change

Advantage of CLR :-



So blank spaces in LR(1) increases which will increase the error detecting capacity
So CLR(1) is more powerful.
The no of states in CLR(1) may be more than any other LR parsers
which will increase blank entries and hence help in error reporting



I4

A→b., a/b

↓

γ3

Canonical collection of LR(1) items

I3 and I6 have same LR(0) but diff lookaheads

I4 and I7 have same LR(0) but diff lookaheads

I8 and I9 have same LR(0) but diff lookahead

CLR(1) parsing table

	a	b	$	S	A
0	S_3	S_4		1	2
1			acc		
2	S_6	S_7			5
3	S_3	S_4			8
4	γ_3	γ_3			
5			γ_1		
6	S_6	S_7			9
7			γ_3		
8	γ_2	γ_2			
9			γ_2		

LALR (1) parsing table

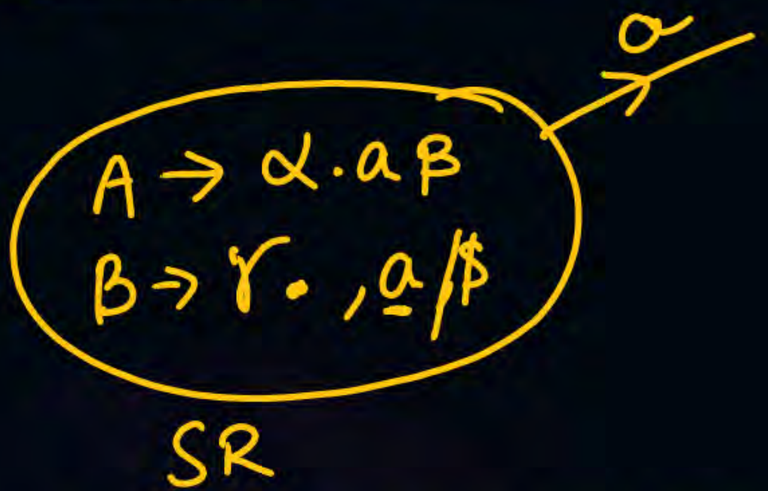
	a	b	$	S	A
0	S_{36}	S_{47}		1	2
1			acc		
2	S_{36}	S_{47}			5
36	S_{36}	S_{47}			89
47	γ_3	γ_3	γ_3		
5			γ_1		
89	γ_2	γ_2	γ_2		

$I_3, I_6 \Rightarrow I_{36}$

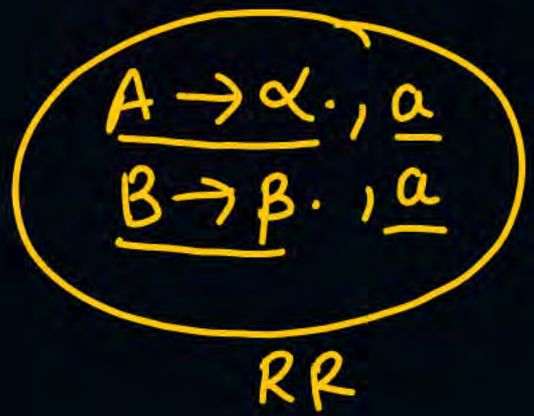
$I_4, I_7 \Rightarrow I_{47}$

$I_8, I_9 \Rightarrow I_{89}$

Conflicts in LR(1) items :



SR

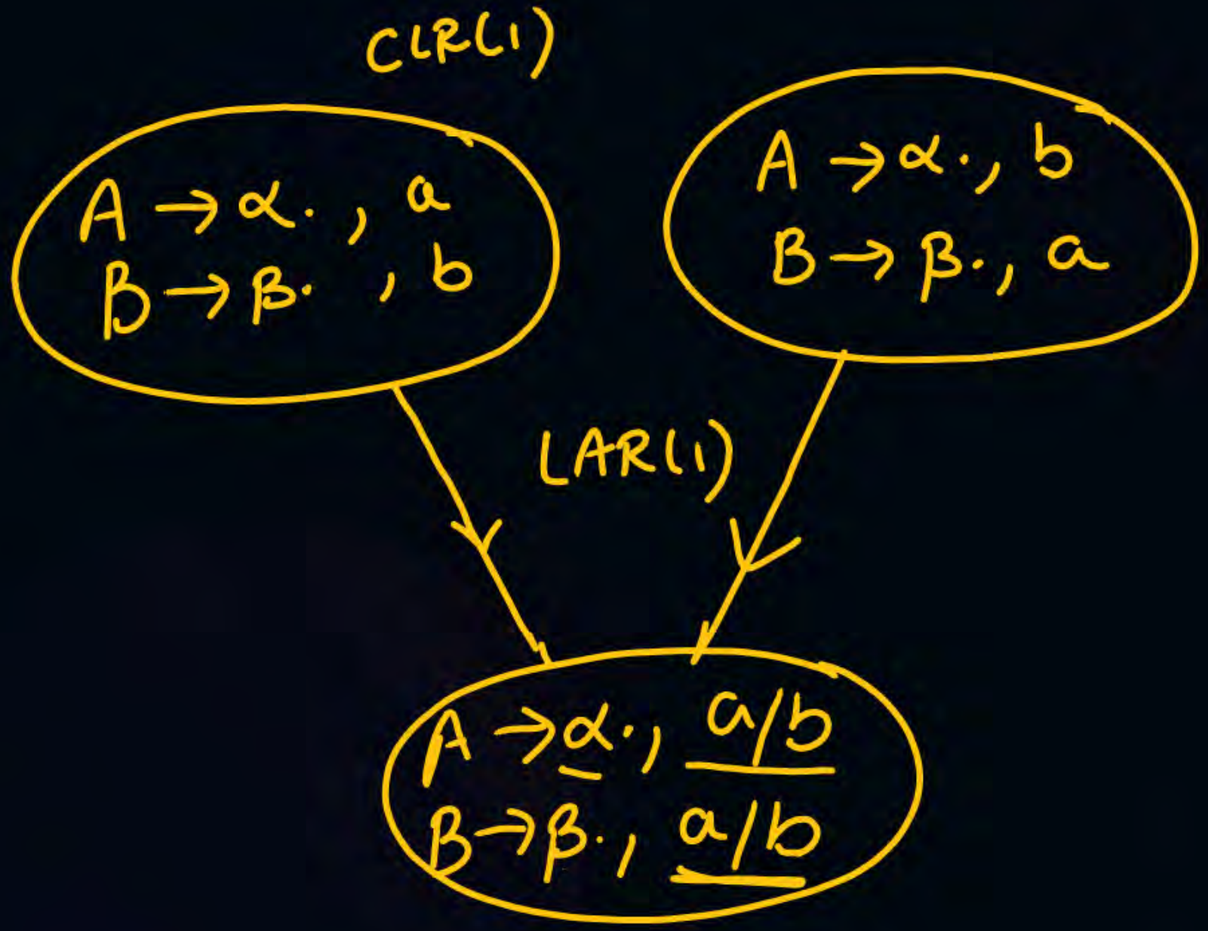


RR

If above conflicts are present, then the grammar is not CLR(1)

If a grammar is not CLR(1), then it is not LALR(1).

Some grammar can be CLR(1) but not LALR(1)


CLR(1)
A → α. , a
B → β. , b
A → α. , b
B → β. , a
LAR(1)
A → α. , a/b
B → β. , a/b

If there are no SR Conflicts in CLR(1), there will be no SR Conflicts in LALR(1)

If there are no RR Conflicts in CLR(1), then may be RR Conflicts in LALR(1)

$S \rightarrow AaAb$ ✓
$\quad / BbBa$ ✓
$A \rightarrow \epsilon$ ✓
$B \rightarrow \epsilon$ ✓

LL(1) ✓

$S' \rightarrow .S$
$S \rightarrow .AaAb$
$\quad / .BbBa$
$A \rightarrow . \rightarrow \{a, b\}$
$B \rightarrow . \rightarrow \{a, b\}$

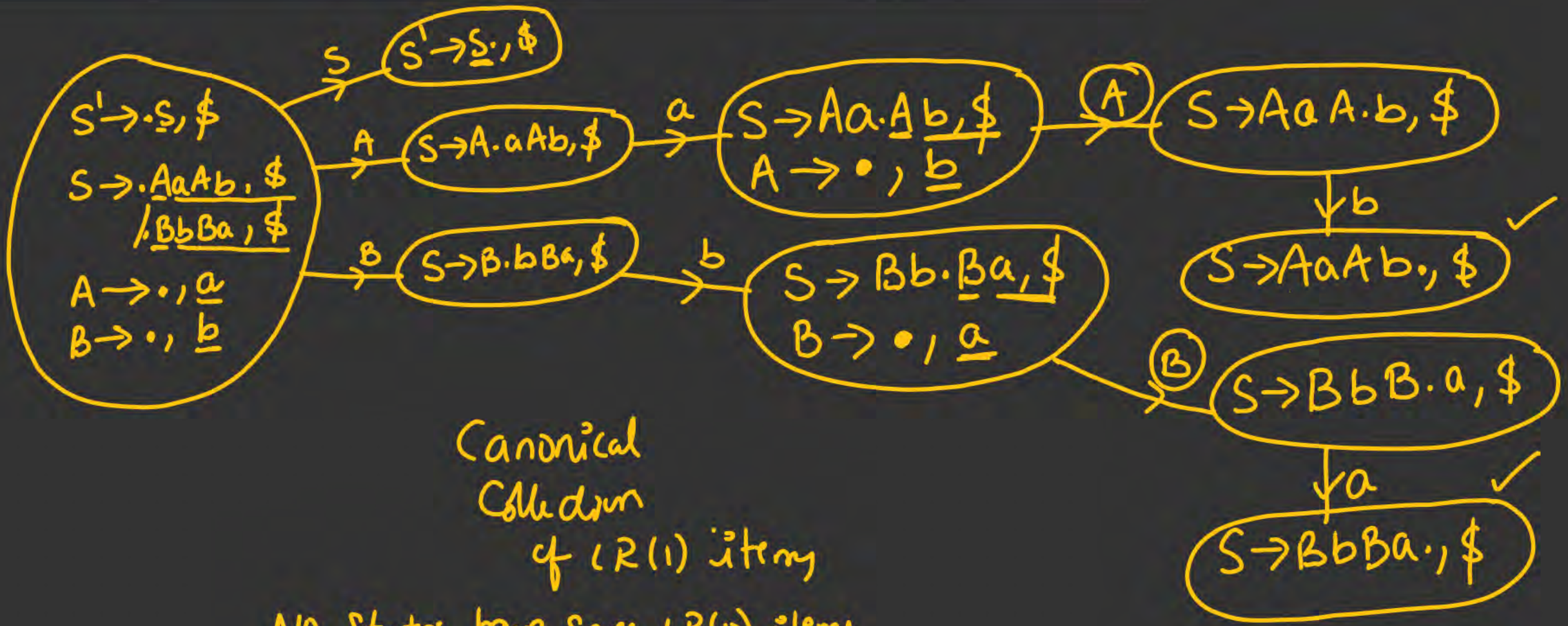
SLR(1) X

NOT LR(0)

CLR(1) ✓
LALR(1) ✓

	a	b
S	$S \rightarrow AaAb$	$S \rightarrow BbBa$

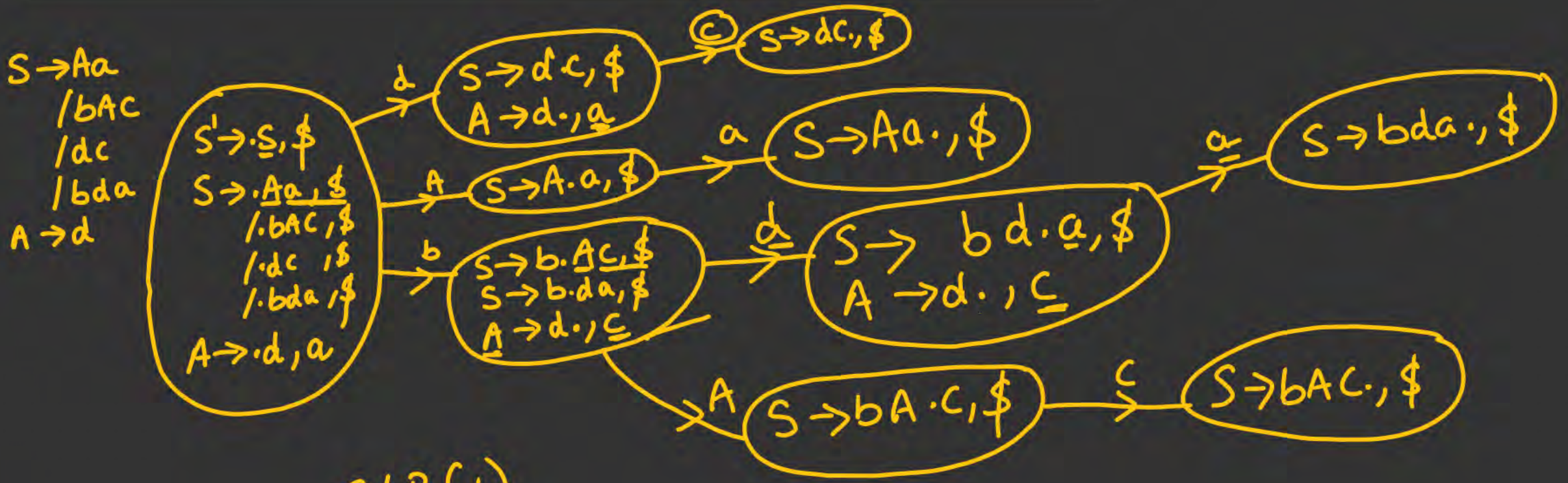
10 min



Canonical Collection of LR(1) items

NO States have same LR(0) items.

So CLR(1) parsing table = LALR(1) parsing table



CLR(1)

No two states have same LR(0) item

So no merging of states

CLR(1) = LALR(1)

$S \to Aa$
$\quad / \underline{bAc}$
$\quad / Bc$
$\quad / \underline{bBa}$
$A \to d$
$B \to d$

$S' \to .S$
$S \to .Aa$
$\quad / .bAc$
$\quad / .Bc$
$\quad / .bBa$
$A \to .d$
$B \to .d$

CLR(1) ✓
LALR(1) ✓

$d \to$ $A \to d.$, $B \to d.$

$A \to d.$ → {a, c}
$B \to d.$ → {a, c}

XSLR(1)

Not LR(0) X

LL(1)

LL(1) X ✓

	b
S	$S \to bAC$ / $S \to bBa$

THANK - YOU